

ECON 21030: Honors Econometrics

Causal Inference 101

Spring Quarter 2024

Causality and causal effects

So far, we have been concerned with modeling and estimating the relationships between different random variables viewed as purely statistical objects. Now we would like to move on to understanding when we can make *causal claims*. If we regress income on education, for example, we might find that people with more education have higher incomes. Is this because people with more education are “smarter” and would have higher earnings whether or not they got more schooling? Or is it because education caused them to have higher earnings, meaning that if they had completed less schooling their expected earnings would have been lower?

Reasoning about causal effects requires reasoning about **counterfactuals**—that is, outcomes in some alternative, imaginary world where something was changed. One approach for doing so from statistics, attributed originally to Jerzy Neyman in the 1920s and that has become increasingly popular in econometrics, is to model counterfactuals using **potential outcomes**. Potential outcomes are random variables directly indexed by the state of the world. In our education example, we might let $Y_i(1)$ be the income of individual i in the world where they go to college, and $Y_i(0)$ be their income in a world where they do not. The causal effect of college attendance for individual i is the difference, or $\tau_i = Y_i(1) - Y_i(0)$. Because τ_i may differ for every individual i , this world is described as a **heterogeneous treatment effects** world. When $\tau_i = \tau \forall i$, then the model is said to exhibit **constant treatment effects**.

We typically think of these potential outcomes as *fixed characteristics* of people: you have an $Y_i(1)$ and $Y_i(0)$ that exist somewhere in the ethereal plane. For any given population of people, you may think of $Y_i(1)$ and $Y_i(0)$ as random variables with their own joint and marginal distributions. In a small population these may be discrete distributions. In a very large population it may be more appropriate to think of them as continuous, depending on what Y_i represents.

The thing that indexes the potential outcome—in this case the binary indicator for college attendance—is known as the **treatment**. It is common to denote treatment, itself a random variable, using D_i or T_i . Because treatment is a random variable, an individual who goes to college has a treatment realization that implies $D_i = 1$. Sometimes people prefer to put treatments in superscripts of potential outcomes, i.e., Y_i^1 and Y_i^0 , which helps to reinforce the idea that these are fixed attributes of i .

The “fundamental problem of causal inference” (Holland 1986) is of course that every individual can experience only *one* treatment. Hence **realized outcomes** are $Y_i = Y_i(1)D_i - Y_i(0)(1 - D_i)$. This makes it impossible to learn about τ_i itself without strong assumptions, because one potential

outcome is always missing. To account for this, we typically move the goal posts slightly and attempt to estimate averages of τ_i in some population. For example, we may seek to estimate the [average treatment effect](#) given by $E[\tau_i]$, which is possible even though individual-level treatment effects τ_i cannot be estimated.¹ Other averages are also interesting, such as the [conditional average treatment effect](#) ($E[\tau_i|X_i]$) or the [treatment effect on the treated](#) ($E[\tau_i|D_i = 1]$). We will discuss how one can estimate such objects shortly.

This may seem like a very general set-up. It embeds a few important restrictions in subtle ways, however. First, notice that potential outcomes depend only on D_i . If D_i is a binary indicator for college attendance, we are taking a very coarse view of what determines Y , abstracting from years of attendance, course load and quality, etc. Usually there is a way to define D_i in a way where this makes sense. We can define D_i as an indicator for *enrolling* in college, for example, and let one treated potential outcome we might be interested in $Y_i(1)$ be the number of years attended or an indicator for obtaining a degree. We can also of course examine multi-valued or continuous D_i . If D_i is years of education, for example, we can have $(Y_i(0), Y_i(1), \dots, Y_i(k))$ be the set of potential outcomes for k possible years of schooling.

Potential outcomes as we have written them also impose a [stable unit treatment value assumption](#) (SUTVA), which implies that individual i 's potential outcomes depend on individual i 's treatment status alone, and not anyone else's. That is, all that matters is my value of D_i , not the value of my friends' or colleagues' D_i s. This rules out interaction and congestion effects, which are plausible in some settings. This too can be relaxed in some ways by changing the definition of i . If we think there are spillovers among units living in the same geographic area, for example, we might measure average outcomes for the area and consider area-level potential outcomes $\bar{Y}_j(\bar{D}_j)$ indexed by the share of individuals treated in the area. The relevant SUTVA assumption then becomes that there is “no interference” between geographic areas, but interactions within a geographic area are unrestricted.

Identifying causal effects

The simplest and most powerful assumption for identifying averages of causal effects is the [independence](#) or [unconfoundedness](#) assumption, which states that:

$$(Y_i(1), Y_i(0)) \perp\!\!\!\perp D_i \tag{1}$$

This assumption implies that the missing potential outcomes are “missing at random”—there are no systematic differences in $Y_i(1)$ or $Y_i(0)$ for people who do and do not receive the treatment. Often this is achieved by explicitly randomizing D_i to ensure it is independent of any other characteristics of individuals, as one would in an RCT.

You can see the power of the unconfoundedness assumption by using our notation for Y_i from

¹You may be thinking it is possible to learn about individual-level treatment effects if we observe the same unit under $D_i = 1$ and $D_i = 0$. That may be true, but requires additional assumptions, such as that potential outcomes are not affected by *prior*. If we consider the actual treatment as the full sequence of past and present treatments, Holland's critique remains valid.

above and considering simple conditional means:

$$\begin{aligned}
\frac{E[Y_i D_i]}{E[D_i]} &= \frac{E[Y_i(1)|D_i = 1]Pr(D_i = 1)}{Pr(D_i = 1)} && \text{by the definition of } Y_i \\
&= E[Y_i(1)] && \text{by independence of } Y_i(1) \text{ and } D_i \\
\frac{E[Y_i(1 - D_i)]}{E[(1 - D_i)]} &= E[Y_i(0)] \\
\frac{E[Y_i D_i]}{E[D_i]} - \frac{E[Y_i(1 - D_i)]}{E[(1 - D_i)]} &= E[Y_i(1)] - E[Y_i(0)] \\
&= E[\tau_i]
\end{aligned}$$

Thus the difference in the conditional means of Y_i given $D_i = 1$ and $D_i = 0$, all of which are directly estimable from their sample analogues, is the average treatment effect.

It is also very common to assume that only [conditional unconfoundedness](#) holds. This means that:

$$(Y_i(1), Y_i(0)) \perp\!\!\!\perp D_i | X_i \quad (2)$$

Here, we are not assuming that D_i is independent of potential outcomes on its own. Instead, treatment is only independent after we control for other factors, which could include observable characteristics such as age, sex, etc., or anything else we measure about individual i . It should be clear that when Assumption 2 holds we can easily estimate the conditional average treatment effect as the conditional difference in means:

$$\frac{E[Y_i D_i | X_i = x]}{E[D_i | X_i = x]} - \frac{E[Y_i(1 - D_i) | X_i = x]}{E[(1 - D_i) | X_i = x]} = E[\tau_i | X_i = x] = \bar{\tau}(x)$$

The overall average treatment effect is just the weighted average of the conditional average treatment effects:

$$E[\tau_i] = E[E[\tau_i | X_i]] = \sum_{x \in X} \bar{\tau}(x) Pr(X_i = x)$$

where I have assumed that the support of X is discrete. It's easy to construct an estimator of this object using the sample analogues of these objects.

Conditional unconfoundedness says that we need to condition on all of X_i in order to estimate causal effects. That turns out to be overkill. The [propensity score theorem](#) says that when $(Y_i(1), Y_i(0)) \perp\!\!\!\perp D_i | X_i$ it is also true that $(Y_i(1), Y_i(0)) \perp\!\!\!\perp D_i | e(X_i)$, where $e(X_i) = Pr(D_i = 1 | X_i)$ is the propensity score (propensity for “propensity to be treated”). This means that it is sufficient to condition on the conditional probability of treatment. We do not need to distinguish between different values of X_i that have the same $e(X_i)$.

The proof of this theorem is fairly straightforward. It shows that $Pr(D_i = 1 | Y_i(1), Y_i(0), e(X_i)) = Pr(D_i = 1 | e(X_i)) = e(X_i)$, which implies that $(Y_i(1), Y_i(0)) \perp\!\!\!\perp D_i | e(X_i)$.

$$\begin{aligned}
E[D_i | Y_i(1), Y_i(0), e(X_i)] &= E[E[D_i | Y_i(1), Y_i(0), X_i] | Y_i(1), Y_i(0), e(X_i)] \\
&= E[E[D_i | X_i] | Y_i(1), Y_i(0), e(X_i)] \\
&= E[e(X_i) | Y_i(1), Y_i(0), e(X_i)] = e(X_i)
\end{aligned}$$

This is powerful result. It implies that we can massively reduce the relevant dimension of X_i we need to condition on to a single scalar, $e(X_i)$. One way to think about the intuition is the following. Suppose there are two values of X_i , x and x' , such that $e(x) = e(x') = p$. Then:

$$\begin{aligned}
E[Y_i|D_i = 1, e(X_i) = p] &= E[Y_i|X_i = x, D_i = 1]Pr(X_i = x|D_i = 1, e(X_i) = p) \\
&\quad + E[Y_i|X_i = x', D_i = 1]Pr(X_i = x'|D_i = 1, e(X_i) = p) \\
&= E[Y_i|X_i = x, D_i = 1] \frac{Pr(D_i = 1|X_i = x, e(X_i) = p)Pr(X_i = x|e(X_i) = p)}{Pr(D_i = 1|e(X_i) = p)} \\
&\quad + E[Y_i|X_i = x', D_i = 1] \frac{Pr(D_i = 1|X_i = x', e(X_i) = p)Pr(X_i = x'|e(X_i) = p)}{Pr(D_i = 1|e(X_i) = p)} \\
&= E[Y_i|X_i = x, D_i = 1]Pr(X_i = x|e(X_i) = p) \\
&\quad + E[Y_i|X_i = x', D_i = 1]Pr(X_i = x'|e(X_i) = p)
\end{aligned}$$

Thus, when we consider the difference in means $E[Y_i|D_i = 1, e(X_i) = p] - E[Y_i|D_i = 0, e(X_i) = p]$, we will recover the weighted average of conditional differences in means $E[Y_i|D_i = 1, X_i = x] - E[Y_i|D_i = 0, X_i = x]$ for all x such that $e(x) = p$. The propensity score insures that there is no imbalance, with some values of X overrepresented among units with $D_i = 1$ or $D_i = 0$. Note that it is still possible that $\bar{\tau}(x) \neq \bar{\tau}(x')$ even when $e(x) = e(x')$.

Propensity scores are used frequently in estimation of causal effects, including in matching and regression frameworks that we will cover below. Regardless of whether you condition on X_i or $e(X_i)$, however, a key requirement for identification of causal effects under conditional unconfoundedness is [overlap](#). If there are some values of X_i where $Pr(D_i = 1|X_i) = 0$ or $Pr(D_i = 1|X_i) = 1$, then identification of causal effects will be problematic because $E[D_i|X_i]$ or $E[1 - D_i|X_i]$ will be zero, meaning that $\bar{\tau}(x)$ is not identified by conditional differences in means. This is an important practical problem for many approaches that use either assumption. Researchers often “trim” the data to the values of $e(X_i)$ or X_i itself such that overlap is gauranteed.

Connections to regression

Under unconfoundedness, average treatment effects are identified from differences in means. This is easy to map onto regression, since we already know how to use them to estimate conditional means. In particular, if Assumption 1 holds, then we know that the parameters of the population regression of Y_i on D_i :

$$Y_i = \alpha + \beta D_i + U_i$$

are given by $\alpha = E[Y_i(0)]$ and $\beta = E[Y_i(1) - Y_i(0)]$. If we have an iid sample (Y_i, D_i) then I can consistently estimate α and β using OLS, and I can construct asymptotically valid standard errors using the usual techniques.

Notice that in this regression U_i has a special meaning. It is *not* simply the residual from a linear projection of Y_i on D_i . It is also:

$$U_i = Y_i(0) - E[Y_i(0)] + D_i(Y_i(1) - Y_i(0) - E[Y_i(1) - Y_i(0)])$$

To distinguish this model from the projection, where U_i was defined by an accounting identity, we call this regression a [causal model](#) or [causal regression](#). This can be a little confusing at first,

since the equation $Y_i = \alpha + \beta D_i + U_i$ looks *a lot* like the projection equations we've been studying so far. Only unconfoundedness gives it special meaning related to average treatment effects.

To see this, imagine running this regression without assuming unconfoundedness holds. I of course can always estimate the regression of Y_i on D_i . But you can show that this would deliver:

$$\begin{aligned}\beta &= \frac{\text{Cov}(Y_i, D_i)}{\text{Var}(D_i)} = \frac{E[Y_i|D_i = 1]Pr(D_i = 1) - E[Y_i]Pr(D_i = 1)}{Pr(D_i = 1)Pr(D_i = 0)} \\ &= E[Y_i(1)|D_i = 1] - E[Y_i(0)|D_i = 0] \\ &= \underbrace{E[Y_i(1) - Y_i(0)|D_i = 1]}_{\text{treatment on the treated}} + \underbrace{E[Y_i(0)|D_i = 1] - E[Y_i(0)|D_i = 0]}_{\text{selection bias}}\end{aligned}$$

The unconfoundedness assumption delivers to things. First, the bias term $E[Y_i(0)|D_i = 1] - E[Y_i(0)|D_i = 0]$ is equal to zero. Secondly, the treatment term $E[Y_i(1) - Y_i(0)|D_i = 1] = E[Y_i(1) - Y_i(0)]$, the average treatment effect. As written, this is the [treatment effect on the treated](#)—i.e., the causal effect of D_i on the population with $D_i = 1$. Note that this decomposition is not unique. We could have also written:

$$\beta = \underbrace{E[Y_i(1) - Y_i(0)|D_i = 0]}_{\text{treatment on the untreated}} + \underbrace{E[Y_i(1)|D_i = 1] - E[Y_i(1)|D_i = 0]}_{\text{selection bias}}$$

which decomposes β into the [treatment on the untreated](#) and a selection bias term. Either way, the regression is only informative about average treatment effects under unconfoundedness.

The preceding discussion suggests it is possible estimate causal effects under weaker assumptions than complete unconfoundedness. Suppose, for example, we had only assumed that $E[Y_i(0)|D_i = 1] = E[Y_i(0)|D_i = 0]$. This implies that D_i is only independent of $Y_i(0)$. Then it is clear from the decomposition formulas above that a regression of Y_i on D_i will recover:

$$\beta_{TOT} = \underbrace{E[Y_i(1) - Y_i(0)|D_i = 1]}_{\text{treatment on the treated}}$$

The structural interpretation of this assumption is that we require only $E[D_i U_i] = E[U_i]$. Note that we do not get the average treatment effect, so though this requirement is enough to recover treatment on the treated it is not enough to recover the parameters of the structural model, namely the average treatment effect $\beta = E[Y_i(1) - Y_i(0)]$. Likewise, I might have assumed that $E[Y_i(1)|D_i = 1] = E[Y_i(1)|D_i = 0]$, implying that β gives me the average treatment effect on the untreated.

These ideas might seem a little strange—why would we be willing to assume $E[Y_i(0)|D_i = 1] = E[Y_i(0)|D_i = 0]$ and not full unconfoundedness? Keep in mind that both assumptions are strictly *weaker* than unconfoundedness, meaning that they are implied by unconfoundedness but not the other way around. Whether they are plausible is going to depend on the setting, question, and model. We'll get into some specific examples later.

What can we recover with regression under conditional unconfoundedness? Let's first consider the case where X_i is discrete and the regression is saturated, so that the controls are dummies $W_i = (1\{X_i = 1\}, \dots, 1\{X_i = K\})'$ for the K points of support in X . Then we might be tempted to “control” for W_i and estimate the effect of the treatment as:

$$Y_i = \beta D_i + W_i' \gamma + U_i$$

To understand what is estimated by β , consider the residualized regression of $\tilde{Y}_i$ on $\tilde{D}_i$. Because W_i is saturated, we know regressing Y_i and D_i on W_i recovers the respective CEFs. It follows that:

$$\begin{aligned}
\beta &= \frac{Cov(\tilde{Y}_i, \tilde{D}_i)}{Var(\tilde{D}_i)} = \frac{E[(Y_i - E[Y_i|X_i])(D_i - E[D_i|X_i])]}{E[(D_i - E[D_i|X_i])^2]} \\
&= \frac{E[Y_i(D_i - E[D_i|X_i])]}{E[(D_i - E[D_i|X_i])^2]} - \frac{E[E[Y_i|X_i](D_i - E[D_i|X_i])]}{E[(D_i - E[D_i|X_i])^2]} \\
&= \frac{E[E[Y_i|X_i, D_i](D_i - E[D_i|X_i])]}{E[(D_i - E[D_i|X_i])^2]} \\
&= \frac{E[(E[Y_i|X_i, D_i = 0] + \bar{\tau}(X_i)D_i)(D_i - E[D_i|X_i])]}{E[(D_i - E[D_i|X_i])^2]} \\
&= \frac{E[E[Y_i|X_i, D_i = 0](D_i - E[D_i|X_i])]}{E[(D_i - E[D_i|X_i])^2]} + \frac{E[\bar{\tau}(X_i)D_i(D_i - E[D_i|X_i])]}{E[(D_i - E[D_i|X_i])^2]} \\
&= \frac{E[\bar{\tau}(X_i)D_i(D_i - E[D_i|X_i])]}{E[(D_i - E[D_i|X_i])^2]} \\
&= \frac{E[\bar{\tau}(X_i)E[D_i(D_i - E[D_i|X_i])|X_i]]}{E[E[(D_i - E[D_i|X_i])^2|X_i]]} \\
&= E[\bar{\tau}(X_i)w(X_i)], \quad w(X_i) = \sigma_D^2(X_i)/E[\sigma_D^2(X_i)]
\end{aligned}$$

where $\sigma_D^2(X_i)$ is the conditional variance of D_i given X_i . This means that OLS gives us a weighted average of the conditional average treatment effects, with weights given by the relative conditional variance of D_i . Points in the support of X_i where D_i is more variable get more weight. Intuitively, those are cells where the *asymptotic variance* of the conditional average treatment effect is smallest. So OLS is putting more weight on the conditional averages that are easiest to estimate.

Is this a good thing? Maybe. We get a weighted average of conditional average treatment effects. And in some settings this average will be exactly equal to the unconditional average treatment effect. Specifically, when $\bar{\tau}(X_i) = \bar{\tau}$, so that there is no heterogeneity in the average treatment effect, then OLS will recover the average treatment effect under Assumption 2. Likewise, if $\sigma_D^2(X_i) = \sigma_D^2$, then OLS will recover the average treatment effect. There are some trivial ways to ensure the latter assumption holds. For example, if $Pr(D_i = 1|X_i) = Pr(D_i = 1)$, then $\sigma_D^2(X_i)$ will naturally not depend on X_i . I leave it as an **exercise** to show that in this case, however, conditional unconfoundedness implies unconditional unconfoundedness, so that we do not need to condition on X_i at all.

As you might have guessed, it is also sufficient to control for propensity scores instead of covariates directly due to the propensity score theorem. A simple way to see why is to note that due to FWL:

$$\beta = \frac{Cov(Y_i, \tilde{D}_i)}{Var(\tilde{D}_i)}$$

But when W_i is saturated $\tilde{D}_i = D_i - E[D_i|X_i] = D_i - e(X_i)$. Regressing D_i on the propensity score will produce a coefficient of 1 (**exercise: why?**). So including the propensity score as regressor in place of W_i will thus produce the same residualized $\tilde{D}_i$ and the same β .

Even though the particular parameter estimated by OLS under conditional unconfoundedness is hard to motivate, it is still the dominant estimand in a lot of applied literature. This is a little

mysterious, since there are simple ways to estimate the ATE using regression as well. In your third homework, for example, you will show that β in the following regression corresponds to the ATE:

$$Y_i = \beta D_i + W_i' \gamma_1 + D_i(W_i - E[W_i])' \gamma_2 + U_i$$

Structural equations

Another way to think about causality is to start with a [structural model](#) or [structural equation](#) directly by writing:

$$Y_i = \alpha + \beta D_i + U_i$$

This is a description of how Y_i is generated—that is, it is a description of the [data generating process](#). The parameters of this model— α and β —need to be defined in order for the description of the model to be complete. Unlike in a regression, they are not going to be defined by the statistical properties of Y_i and D_i .

For example, the simplest way to do so is to consider a specific implementation of the potential outcomes framework we have been working with so far. We can define $\alpha = E[Y_i(0)]$, $\beta = Y_i(1) - Y_i(0)$, and $U_i = Y_i(0) - E[Y_i(0)]$. Here, U_i is a [structural residual](#) that governs population heterogeneity in $Y_i(0)$, and treatment effects are constant for all i and equal to β . Again, these definitions are *part* of the structural model: we are giving special meaning to β , α , and U_i .

We might also wish to write down a model that allows for heterogeneous treatment effects as:

$$Y_i = \alpha + \beta_i D_i + U_i$$

You can see that this model fully nests our potential outcomes model if we define:

$$\begin{aligned}\alpha &= E[Y_i(0)] \\ \beta_i &= Y_i(1) - Y_i(0) \\ U_i &= Y_i(0) - E[Y_i(0)]\end{aligned}$$

Average treatment effects are given by: $E[Y_i(1)] - E[Y_i(0)] = E[\beta_i]$.

Regardless of the definitions of α , β and U_i , if we observed (Y_i, D_i, U_i) it would of course be trivial to estimate α and β . One could simply compute the mean of $Y_i - U_i$ given $D_i = 0$ to get α , and use mean of $Y_i - U_i$ given $D_i = 1$ to get β (or the average of β_i in the heterogeneous case). The problem is that the structural residual U_i is an [unobservable](#), so this strategy is infeasible.

Instead, the data only contain information on Y_i and D_i . If we want to use a population regression of Y_i on D_i to identify β , we need to make an additional assumption *NOT* (yet) imposed by the structural model: that $E[D_i U_i] = 0$ ($E[U_i] = 0$ holds by definition). This is saying something substantive about the correlation between the random variables D_i and U_i . In the constant effect model, for example, it requires that treatment is uncorrelated with $Y_i(0)$, meaning people do not self-select on the “level” of untreated potential outcomes. In other words, in this model assuming that $D_i U_i = 0$ implies (and is implied by) a type of unconfoundedness.

In the heterogeneous effect model where $\beta_i = Y_i(1) - Y_i(0)$, $D_i U_i = 0$ requires that those who take up treatment are not select on levels as well. However, it remains possible that β_i and D_i are correlated, allowing for some selection on “gains.” If we imposed $E[\beta_i D_i] = E[\beta_i]E[D_i]$ we would also be assuming that those who take up treatment are no more (or less) likely to benefit from it, on average.

When these restrictions hold, we say that β , the parameter of the structural model, is **identified** by a regression of Y_i on D_i . This argument is the source of the common confusion that regression estimates causal effects when “treatment is uncorrelated with the residuals.” But this is a completely vapid thing to say—regression residuals are uncorrelated with covariates *by construction*. Instead, we require that treatment is uncorrelated with *structural residuals*. That is a statement about unobserved factors that is not directly testable without further assumptions, nor is it guaranteed to hold in your data due to the mechanical properties of regression.

We might also write down structural models that embed a conditional independence restriction. Suppose, for example, that my structural model is:

$$Y_i = \alpha + \beta D_i + X_i' \gamma + U_i \quad (3)$$

As before, here we might define:

$$\begin{aligned} E[Y_i(0)|X_i] &= \alpha + X_i' \gamma \\ \beta &= Y_i(1) - Y_i(0) \\ U_i &= Y_i(0) - E[Y_i(0)|X_i] \end{aligned} \quad (4)$$

You can see immediately how incorporating X_i into the model allow for more flexibility. We now have a model for the conditional mean of $Y_i(0)$ given X_i , and U_i is the deviation from those conditional means. The assumption that $E[Y_i(0)|X_i] = \alpha + X_i' \gamma$ is also potentially making a restriction on the conditional expectation function for $Y_i(0)$. If X_i is not saturated, for example, this restriction may embed additional structure, such as requiring the conditional mean of $Y_i(0)$ to be linear in some X_i terms.

This model also makes clear how to incorporate omitted variable bias type logic into reasoning about causal effects. Suppose I believe that the true model is described in Equations 3 and 4 and that $E[D_i U_i] = 0$, so that if I observed the joint distribution (Y_i, D_i, X_i) I could directly identify β using regression. If instead we only observe some subset of X_i , say W_i and estimate

$$Y_i = \tilde{\alpha} + \tilde{\beta} D_i + W_i' \tilde{\gamma} + V_i$$

under the assumption that $E[V_i D_i] = 0$, the relationship between β (what we want to know) and $\tilde{\beta}$ (what we can identify) will depend on how D_i correlates with the components of X_i omitted from the model and the coefficients on those components in Equation 3. Sometimes theory or intuition will guide you about the potential direction of that bias. We will think of some examples in class based on recent work.

Finally, note that we can also incorporate more heterogeneity by writing:

$$Y_i = \alpha + \beta_i D_i + X_i' \gamma + U_i$$

where β_i allows treatment effects to be heterogeneous. As in the unconditional case, we need to make additional assumptions about how D_i , U_i , and β_i correlate in order to be able to identify β or averages of β_i from the distribution of Y_i and D_i .

Alternative types of structural equations

Of course, structural equations don't need to be motivated by a potential outcomes framework. Suppose, for example, I wrote down a model for supply and demand that says:

$$\begin{aligned}\log(Q^d) &= \beta \log(P) + U \\ \log(Q^s) &= \gamma \log(P) + V\end{aligned}$$

where U and V are random variables that shift demand and supply, respectively. U and V might sometimes be called in your macro textbooks demand and supply “shocks.” Equilibrium is determined by the intersection of supply and demand, so that the quantity and price observed in the market is given by, e.g.,

$$\begin{aligned}\log(P) &= \frac{U - V}{\gamma - \beta} \\ \log(Q) &= \beta \log(P) + U = \gamma \log(P) + V\end{aligned}$$

When would a regression of $\log(Q)$ on $\log(P)$ recover the elasticity of demand β ?

This simple example may seem very different from the potential outcomes framework we've been studying so far. But that is mostly an artifact of notation. Both are attempting to make statements about counterfactuals, e.g., how much would demand change in response to a 10% price increase? We could of course write Q^d itself as a potential outcome indexed by P , e.g., $Q^d(P)$. And likewise, market-level quantities might be indexed by U and V , e.g., $Q(U, V)$. The point again is that the parameter is defined by the model and how it is identified (or not) depends on our assumptions how potentially unobserved factors are or are not correlated with observed factors.

As another example, suppose we observe output Y_i across a series of firms i , perhaps in a given industry. We also observe each firm's labor and capital input, L_i and K_i . We are interested in estimating the production elasticities which tell us how a given firm's output increases if it were to add more labor or capital to the mix. Note that these are, like with the examples above, causal parameters: we want to know the average effects of L_i and K_i on Y_i .

We might next assume a particular production function. For example, we might model $Y_i = A_i L_i^\alpha K_i^\beta$ for some parameters α and β and some unobserved productivity shocks A_i . This is a Cobb-Douglas production function, which you may have seen in previous classes. It assumes that labor and capital are combined with constant elasticities (α, β) , simplifying our objective to estimating these two numbers. In this model, log output for firms can be written

$$\log(Y_i) = \alpha \log(L_i) + \beta \log(K_i) + \log(A_i)$$

Now suppose we, as the researchers, only observe the joint distribution of Y_i, L_i and K_i . In this case average output among firms with input choices (L_i, K_i) is simply:

$$E[\log(Y_i)|L_i, K_i] = \alpha \log(L_i) + \beta \log(K_i) + E[\log(A_i)|L_i, K_i]$$

You can see immediately that $E[\log(A_i)|L_i, K_i]$ functions as a kind of regression residual. If we want to use regression to estimate α and β , we're going to need extra assumptions. In particular, we

will need to assume that firms make labor and capital investment decisions prior to the realization of productivity A_i , which is distributed identically and independently across firms. Formally, this implies A_i is independent of (L_i, K_i) , so that $E[\log(A_i)|L_i, K_i] = E[\log(A_i)]$. Thus:

$$E[\log(Y_i)|L_i, K_i] = \gamma + \alpha E\log(L_i) + \beta \log(K_i)$$

where $\gamma = E[\log(A_i)]$. Under these assumptions, the CEF of $\log(Y_i)$ is linear in $\log(L_i)$ and $\log(K_i)$ (and a constant), and the slopes on L_i and K_i are equal to the parameters of the production function. We can therefore identify α and β from a regression of $\log(Y_i)$ onto $\log(L_i)$ and $\log(K_i)$.

Research designs

Doing applied research in economics typically requires two processes: Defining the parameters of interest and developing a [research design](#). The first part of this process we have already discussed at length. One needs to write down a model and define the parameters you care about. These may be simple causal effects or, as just illustrated, the parameters of more complicated economic models. The second part requires writing down a set of assumption that, combined with the random variables whose joint distribution you can observe, allows you to identify and estimate those parameters. That is, it allows you to define an estimand that can be mapped to the parameters of interest.

Sometimes there are multiple research designs one might consider. For example, if we are interested in the following structural equation

$$\begin{aligned} Y_i &= \alpha + \beta D_i + X_i' \gamma + U_i \\ E[Y_i(0)|X_i] &= \alpha + X_i' \gamma \\ \beta &= Y_i(1) - Y_i(0) \\ U_i &= Y_i(0) - E[Y_i(0)|X_i] \end{aligned}$$

there are many different choices for what I might include in X_i . If I'm going to use regression to estimate β , then I will also need to impose $E[U_i D_i] = 0$ —i.e., conditional independence. How credible this assumption is depends in part on what I put in X_i and how D_i is generated. If D_i was randomly assigned as part of an experiment, we might be comfortable assuming that $E[U_i D_i] = 0$ even with limited X_i (or none). If D_i reflects the choices of i themselves, we might be skeptical that it is uncorrelated with U_i unless we can observe a lot about i (i.e., a very rich set of X_i).

Suppose, for example, we are trying to understand the causal effects of smoking on lung cancer. Would you believe that the decision to smoke is uncorrelated with individual-level cancer risk, as would be required in a model with no X_i ? If not, what kind of observables X_i would you want to control for before you were comfortable believing that smoking D_i was uncorrelated with the residual unobserved variation in baseline cancer risk $(Y_i(0) - E[Y_i(0)])$?

Sometimes, the assumptions we make will be [testable](#). If we believe D_i was unconditionally randomly assigned for example, then it follows that $E[D_i X_i] = E[D_i] E[X_i]$ for any X_i we observe. One could test such hypotheses with data. In general, however, identifying a causal effect is going to require a combination of modeling and distributional assumptions. It is up to you, as the researcher and consumer of research, to decide how credible you think those assumptions are. And if you think they are not very credible, it's up to you come up with better ones!